

Surrogate Dataset Design Plan

Planning Gate Before Any Dataset Construction

Codex-assisted surrogate planning report

June 17, 2026

Abstract

This report defines the planned schema, parameter grid, label policy, mask policy, diagnostics, split strategy, and construction gate for a future surrogate dataset. It approves dataset design only. It does not generate data, labels, arrays, neural-network files, training tables, broad parameter sweeps, or surrogate models. All future labels must come from the validated baseline American Crank–Nicolson / projected SOR solver unless a later human-review gate approves a separately labelled alternative.

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1 Purpose of the Dataset Design Plan

The application-strength gate concluded `READY_FOR_DATASET_PLANNING_WITH_LIMITATION S`. The next step is therefore a written dataset design plan, not dataset construction. This report records the planned future schema and quality policy so that a later construction ticket can be reviewed before it creates any arrays or labels.

The plan follows the project’s evidence ladder: classical American option validation, pilot stress-map diagnostics, quantitative risk analytics, and then dataset planning. This order is important because a neural surrogate trained on unreliable numerical labels would only learn unreliable structure. The mathematical background remains the Black–Scholes American free-boundary and finite-difference setting (Black and Scholes, 1973; Merton, 1973; Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Link to Solver Validation and Application-Strength Gate

Ticket 12 passed solver validation with limitations. The downstream design gate approved only a small pilot. Pilot 01 and quantitative risk analytics then showed coherent application messages: put boundaries moved lower as volatility increased, dividend-call boundaries moved lower as dividend yield increased, high volatility increased PSOR effort, and LCP diagnostics remained small in the tested regimes.

Those results support dataset planning. They do not support immediate generation, broad sweeps, neural training, production Greek labels, or final paper claims.

Prior gate	Constraint carried into this plan
Ticket 12 synthesis	Solver is validated with limitations, not production certified.
Application-strength gate	Planning is allowed; construction and training remain blocked.
Pilot 01	Use baseline CN/PSOR, $K = 1$, and full diagnostics.
Quantitative analytics	Preserve boundary, Gamma, PSOR, runtime, and LCP metadata.
Reference integration	Use verified citations and avoid unsupported literature claims.

3 Dataset Purpose and Non-Generation Statement

The future dataset is intended to support constrained surrogate experiments for American option price and continuation-premium surfaces, boundary diagnostics, and possibly masked Greek diagnostics. It should help answer whether a surrogate can accelerate scenario evaluation while respecting the payoff obstacle, continuation premium, and boundary-aware risk structure.

This step does not produce the dataset. The files in `reports/05_surrogate/` are planning artifacts only. No `.npz`, `.npy`, `.pt`, training table, model file, or generated label file is created by this design plan.

4 Solver Source and Default Numerical Settings

All future labels must come from the validated baseline solver:

```
solver_name=american_crank_nicolson_psor_price
solver_variant=baseline_cn_psor
```

Rannacher-smoothed outputs are not the default. Ticket 10A kept Rannacher as an optional comparison tool only (Rannacher, 1984). If Rannacher is ever included in a later dataset, it must be separately labelled and must not be pooled silently with baseline CN/PSOR labels.

The first construction plan should inherit the downstream defaults unless human review changes them: normalized strike $K = 1$, default $S_{\max} = 4K$, default $M = N = 120$, and selected higher-grid confirmation at $M = N = 180$ before broad generation.

5 Option Families

The planned dataset must include:

- American puts;
- dividend-paying American calls.

Both families are required because the application analytics produced meaningful but different messages for puts and dividend-paying calls. A pooled dataset that hides dividend-call boundary weakness would not serve the project. No-dividend American calls remain useful as validation controls, but they are not the core first dataset family.

6 Feature Schema

The primary learning features are normalized variables:

$$x = (\log(S/K), \tau/T, r, q, \sigma, T, \text{is_call}).$$

This matches the methodology handout's normalized feature recommendation while keeping enough metadata for solver audit.

Feature	Definition	Required	Notes
log_moneyness	$\log(S/K)$.	yes	Primary spot feature.
tau_fraction	τ/T .	yes	Normalized time-to-maturity.
r	Risk-free rate.	yes	Scale consistently in preprocessing.
q	Continuous dividend yield.	yes	Needed for both option families.

Feature	Definition	Required	Notes
<code>sigma</code>	Volatility.	yes	Stress-sensitive regime feature.
<code>T</code>	Maturity in years.	yes	Planned grid includes four values.
<code>is_call</code>	1 for call, 0 for put.	yes	Numeric option-family indicator.
<code>option_type</code>	put or call.	yes	Human-readable audit field.
<code>S_over_K</code>	S/K .	optional	Redundant but useful for reporting.
<code>Smax, M, N</code>	Solver domain and grid metadata.	yes	Required for reproducibility.

The companion CSV version is listed in the reproducibility and source record.

7 Label Schema

The first design separates reliable price-style labels from diagnostic labels. This protects the project from treating every computed quantity as equally trustworthy.

Label	Definition	Reliability	Allowed use
<code>value_over_K</code>	U/K .	core reliable	Primary future label.
<code>payoff_over_K</code> <code>premium_over_K</code>	Φ/K . $(U - \Phi)/K$.	analytic reliable reliable with caution	Reference label. Primary future label.
<code>exercise_indicator</code>	Threshold-derived exercise flag.	diagnostic	Metadata only at first.
<code>boundary_spot_over_K</code>	Extracted boundary divided by K .	diagnostic	Boundary metadata.
<code>delta</code>	Finite-difference Delta.	cautioned	Optional diagnostic label.
<code>scaled_gamma</code>	$K\Gamma$.	high caution	Masked diagnostic only.

Price and premium labels are the most reliable first targets. Boundary labels remain threshold-based. Gamma labels are fragile near payoff kinks, maturity, and extracted boundaries; they must never be used unmasked. The companion CSV version is listed in the reproducibility and source record.

8 Parameter Grid Plan

The first planned grid is intentionally small:

$$T \in \{0.25, 0.5, 1.0, 2.0\}, \quad \sigma \in \{0.20, 0.40, 0.60\}, \quad r \in \{0.01, 0.05, 0.10\}.$$

For dividend yield:

$$q \in \{0.00, 0.03, 0.06, 0.10\}.$$

For dividend-paying calls, the first construction plan should ensure positive-dividend regimes are included. The $q = 0$ no-dividend call is mainly a validation control and should not dominate the dividend-call dataset.

This grid is a plan, not generated data. The companion CSV version is listed in the reproducibility and source record.

9 Spot-Time Sampling Plan

Future solves should keep the full solver domain for boundary conditions and diagnostics. The main learning and reporting region should be:

$$S/K \in [0.4, 1.8], \quad \tau/T \in [0, 1].$$

The recommended first version is a regular-grid design: retain grid nodes in the target region because they are reproducible and easy to audit. Boundary-near, kink-near, and maturity rows should be marked by masks rather than oversampled in this design step. Random samples from surfaces and boundary-focused oversampling may be considered later, but only after the first construction plan fixes masks, splits, and diagnostics.

10 Mask Policy

Masks are mandatory metadata. They are not cosmetic fields. They tell future surrogate code and human reviewers which labels are safe to use for which purpose.

Mask	Meaning	Required use
payoff_kink_near	Node is near $S = K$, where payoff slope changes.	Greek caution.
boundary_near	Node is near the extracted free-boundary diagnostic.	Greek and boundary caution.
maturity_row	Node belongs to $\tau = 0$.	Greek caution.
strict_interior	Node avoids endpoints, kink, maturity, and boundary bands.	Safer Greek summaries.
gamma_allowed_mask	Node is allowed for Gamma target use.	Required for Gamma.
delta_allowed_mask	Node is allowed for Delta target use.	Required for Delta.
exercise_region	Threshold-derived exercise-like region.	Boundary metadata.

Mask	Meaning	Required use
<code>continuation_region</code>	Threshold-derived continuation-like region.	Boundary metadata.

The mask policy follows the Ticket 09 boundary extraction and Ticket 10 Greek-diagnostic logic. It also follows the handout warning not to train heavily on Gamma exactly at payoff kinks or extracted boundaries.

11 Diagnostic Metadata Policy

Every future solved regime must store diagnostics before any label is accepted:

- all PSOR steps converged;
- maximum and mean PSOR iterations;
- maximum final PSOR update;
- maximum obstacle violation;
- maximum equation violation;
- maximum absolute complementarity product;
- boundary found count and boundary threshold;
- M , N , $S_{\max}$, ΔS , and $\Delta \tau$;
- solver name and solver variant;
- downstream use status.

If a regime fails convergence or diagnostic checks, it must be excluded from label export until human review decides how to handle it. The companion CSV version is listed in the reproducibility and source record.

12 Train/Test Split Policy

The main split must be regime-level: hold out entire parameter combinations. A random point split inside a solved surface is allowed only as a secondary debugging diagnostic. Neighboring grid points from the same surface are highly correlated; using them in both train and test can make a surrogate look better than it really is.

Split type	Purpose	Priority
Regime holdout	Test generalization to unseen parameter combinations.	primary
Option-family balance	Ensure puts and dividend calls are both represented.	primary

Split type	Purpose	Priority
Stress-regime holdout	Test high-volatility, high-dividend, or low-rate regions.	primary
Random point split	Debug interpolation within solved surfaces.	secondary only
Boundary-band holdout	Optional future boundary-specific diagnostic.	review required

The machine-readable version is `reports/05_surrogate/dataset_split_policy.csv`.

13 Storage Recommendation

The future construction ticket may use compressed arrays such as `.npz` plus CSV or JSON manifests. A reasonable future package would include one array file for numeric samples and one human-readable manifest for regimes, diagnostics, masks, and split membership.

This report does not create that storage. No array file, model file, training table, or label export is produced here.

14 Dataset Construction Gate

Dataset construction may start only after human review approves:

- parameter grid;
- feature schema;
- label schema;
- mask policy;
- diagnostic policy;
- split strategy;
- storage format;
- handling of failed diagnostics.

Approval of this design plan is not approval to train models. A later surrogate plan must review model architecture, losses, metrics, held-out regime tests, and PSOR spot-check policy.

15 Blocked Actions

The following remain blocked:

- actual dataset construction;
- `.npz`, `.npy`, `.pt`, or training-table export;

- neural surrogate training;
- broad production-scale sweeps;
- unmasked Gamma labels;
- final paper claims based on a dataset that does not yet exist.

16 Limitations

The planned grid is deliberately small. It should be treated as a first design, not a production coverage claim. Boundary labels are threshold-based. Gamma labels remain fragile. Runtime and PSOR effort may change under larger grids or wider parameter ranges. The plan also does not yet specify exact train/test regime IDs; that should be done in the construction plan before any data is generated.

17 Recommended Next Step

The recommended next step is human review of the five schema and policy CSVs plus this report. If accepted, the next technical artifact should be a dataset construction plan that names the exact regimes, storage format, diagnostic failure rules, and split assignments. Only after that plan is approved should the project generate any dataset files.

18 Reproducibility and Source Record

Item	Record
Feature schema	reports/05_surrogate/dataset_feature_schema.csv.
Label schema	reports/05_surrogate/dataset_label_schema.csv.
Parameter grid plan	reports/05_surrogate/dataset_parameter_grid_plan.csv.
Split policy	reports/05_surrogate/dataset_split_policy.csv.
Diagnostic policy	reports/05_surrogate/dataset_diagnostic_policy.csv.
Backlog	reports/05_surrogate/dataset_generation_backlog.md.
Design report	reports/05_surrogate/surrogate_dataset_design_plan.tex.
Shared bibliography	reports/03_solver/references.bib.

References

Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141, 1973. doi: 10.2307/3003143.
- Rolf Rannacher. Finite element solution of diffusion problems with irregular data. *Numerische Mathematik*, 43(2):309–327, 1984. doi: 10.1007/BF01390130.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.